

Critical Review

Charged Particle Radiation Therapy for Uveal Melanoma: A Systematic Review and Meta-Analysis

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Charged particle therapy (CPT) delivered with either protons, helium ions, or carbon ions, has been used to treat uveal melanoma. The present analysis was performed to systematically evaluate the efficacy and adverse effects of CPT for uveal melanoma. We searched EMBASE, MEDLINE, Cochrane Central Register of Controlled Trials, Cochrane Database of Systematic Reviews, and SciVerse Scopus and cross-referenced recent systematic reviews through January 2012. Two independent reviewers identified clinical trials and observational studies of CPT (protons, helium ions, and carbon ions). These reviewers extracted data and assessed study quality. Twenty-seven studies enrolling 8809 uveal melanoma patients met inclusion criteria. The rate of local recurrence was significantly less with CPT than with brachytherapy (odds ratio [OR] = 0.22, 95% confidence interval [CI], 0.21-0.23). There were no significant differences in mortality or enucleation rates. Results were robust in multiple sensitivity analyses. CPT was also associated with lower retinopathy and cataract formation rates. Data suggest better outcomes may be possible with charged particle therapy with respect to local recurrence, retinopathy, and cataract formation rates. The overall quality of the evidence is low, and higher quality comparative effectiveness studies are needed to provide better evidence. © 2013 Elsevier Inc.

Introduction

Despite the low incidence of uveal melanomas, they represent the most common primary tumor of the eye in adults and can be devastating, with significant morbidity and mortality. Approximately 2200 cases are diagnosed each year in the United States (1, 2). Historically, these tumors were diagnosed clinically, and patients were treated surgically, most often with enucleation. While this was a successful strategy for treating the cancer in many instances, enucleation resulted in loss of the eye and vision. Less invasive options such as radiation therapy have also been used with the hope of eradicating tumor while preserving the eye and useful vision. The radiation therapy options most commonly

used included brachytherapy and CPT. Brachytherapy is the delivery of radiation with the source of irradiation placed a short distance from the tumor. This is accomplished by sewing a plaque containing radioactive sources directly to the eye over the region of the tumor. The high-dose region around the brachytherapy source is quite localized because the dose decreases quickly as the distance increases from the source, with dose proportional to $1/r^2$ (where r = radius or distance from the source to reference point). The plaque is left in place until the prescribed dose is delivered and then it is removed. This requires anesthesia and 2 invasive operative procedures. The radioactive isotopes most commonly used for this purpose are iodine-125, ruthenium-106, and palladium-103. Early studies of brachytherapy reported

Note—An online CME test for this article can be taken at <http://astro.org/MOC>.

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